AWS Cloud Practitioner

This course provides a comprehensive introduction to Amazon Web Services (AWS). It covers the fundamental concepts of cloud computing, core AWS services, security best practices, and cost optimization strategies. By the end of this course, learners will have a solid understanding of AWS and be able to design, deploy, and manage cloud-based applications on the AWS platform.

Course Objectives

Upon completion of this course, learners will be able to:

- Explain the core concepts of cloud computing and the benefits of using AWS
- Identify and describe key AWS services, including EC2, S3, VPC, IAM, and more
- Understand the AWS global infrastructure and regions
- Create and manage AWS resources using the AWS Management Console and CLI
- Implement basic security best practices on AWS
- Optimize AWS costs using various strategies
- Prepare for the AWS Certified Cloud Practitioner exam

Module 1: Introduction to Cloud Computing

- What is cloud computing?
- Benefits of cloud computing
- Types of cloud computing (public, private, hybrid, multi-cloud)
- Key cloud computing concepts (IaaS, PaaS, SaaS)

Module 2: Core AWS Services

- **Security, Identity, and Compliance:** IAM (Identity and Access Management) · Security Groups and Network ACLs · Key Management Service (KMS)
- **Networking and Content Delivery:** VPC (Virtual Private Cloud) · Route 53 · CloudFront
- **Compute Services:** EC2 (Elastic Compute Cloud) — instance types, AMIs, key pairs, security groups · Lambda
- **Storage Services:** S3 (Simple Storage Service) · EBS (Elastic Block Store)
- **Database Services:** RDS (Relational Database Service) · DynamoDB

Module 3: AWS Global Infrastructure

- Regions and Availability Zones
- Edge Locations
- Global Accelerator

Module 4: Billing and Cost Optimization

- Understanding AWS Pricing Models
- Cost Explorer
- Reserved Instances and Savings Plans
- Rightsizing Instances
- Auto-Scaling

Module 5: Best Practices and Troubleshooting

- Best practices for building secure and reliable applications on AWS
- Common troubleshooting techniques
- AWS Well-Architected Framework
- Exam preparation

Hands-on Labs

Module 2 is taught in five short blocks, each followed immediately by its lab, so teaching content always precedes hands-on work. Each lab builds on the previous one: the VPC from Lab 2 hosts the servers from Labs 3–4 and the database from Lab 7, and the IAM role from Lab 1 is used from EC2 in Labs 3 and 5. The final lab ends with a complete cleanup checklist.

Lab	Title	Duration
1	Configuring IAM Roles and Policies	30 min
2	Configuring Security Groups and Network ACLs	30 min
3	Creating EC2 Instances	25 min
4	Deploying a Web Application on EC2	30 min
5	Using S3 for Object Storage	25 min
6	Implementing Serverless Functions with Lambda	25 min
7	Setting Up a Relational Database with RDS	30 min
8	Implementing Cost Optimization Strategies	30 min

Certification Alignment

The course maps to the four domains of the **AWS Certified Cloud Practitioner (CLF-C02)** exam — 65 questions in 90 minutes, passing score 700/1000.

Exam domain	Weight	Covered in
1. Cloud Concepts	24%	Modules 1, 3 and 5
2. Security and Compliance	30%	Module 2 (Security), Labs 1–2, Module 5
3. Cloud Technology and Services	34%	Module 2, Module 3, Labs 2–7
4. Billing, Pricing and Support	12%	Module 4, Lab 8, Module 5